REBLOCK

Turning ceramic waste into high-performance permeable paving block

Contact Information

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Team

Minchan KIM | Founder & CEO

- 3 years of leading tech-based ESG commercialization projects
- Environmental research projects with the Korean Society for Microbiology & Korea Forest Service
- Member, Korea Block Association

Seonghyun PARK | CTO

- Eco-friendly landscaping projects
- Member, Korea Block Association

Dayeon JUNG | COO

- B2B trade sales between Korea and Spain
- UC Berkeley Haas, I-Corps program

Jinsu YUN | CBO

- 2 yrs as CEO of a social enterprise
- Business Administration, Konkuk Univ.

Advisors

- Eco-friendly Concrete Lab, Konkuk Univ.
- Korean Society for Life Cycle Assessment



Recycled ceramic aggregate & REBLOCK block



First pilot installation, 24 m² in Konkuk Univ., Seoul (Jun 2026)

Problem

- **Urbanization seals the ground:** asphalt and concrete seal out rainwater → surface runoff and flash floods rise while groundwater recharge declines
- **Existing permeable blocks fall short:** higher permeability means weaker blocks, and clogging within 2–3 years erodes performance
- **Ceramic waste piles up:** 2.5M t/yr in Korea, mostly landfilled → yet its hardness and reactive minerals are exactly what permeable paving needs

Solution

- **Ceramic waste aggregate:** Crushed and graded to 0~10 mm, replacing **90%+** of general aggregate
- **Waste-blend mix design:** Combined with 3~4 additional waste-derived materials for a stable, low cost formulation
- **Pozzolanic bonding:** Silica and alumina in ceramic waste reinforce the aggregate-cement bond → high permeability without losing strength (patent pending)

Key Features

*KS: Korean Industrial Standard

- **Verified performance:** 4 accredited KS F 4419 performance tests by **KCL** and **KTR** (national testing institutes)
- Permeability **1.12 mm/s** (11× *KS) & flexural strength **4.4 MPa** & Seoul city durability **Grade 2**
- Drop-in replacement: standard 200×200 mm (T60 / T80), 6 colors, existing installation methods

Business Model

- **Product sales: USD 16 /m²** to public agencies and private contractors
- **Waste processing fee: USD 40/t** paid by waste collectors – we get paid to receive our own raw material
- **Circular take-back:** used blocks recovered and remanufactured up to 2 cycles

Target Customers

- **B2G:** national and public agencies (sidewalks, parks, flood-prone zones)
- **B2B:** private contractors, landscape contractors, design firms (spec-in)

Business Snapshot

- **33 t** produced via contract manufacturing: pilot **24 m²** installed (Jun 2026) → 2nd pilot confirmed
- Paid pilot and supply talks underway with 2 Korean municipalities
- **Grand Prize**: 2026 Climate, Energy & Environment Startup Competition (Korean Ministry of Climate, Energy and Environment)
- **Top Prize**: Defense Tech Startup Competition · **Winner**: Enactus Korea 2026 · Future Bridge Competition by Korea Hydro & Nuclear Power Co., LTD
- **Roadmap:** 4 pilots (2026) → certified performance & public procurement (2027) → In-house crushing plant (2028)

Seeking Opportunities

- Pilot partners in Singapore & Southeast Asia: public agencies, contractors, landscape & design firms
- Local partners for ceramic waste sourcing and localized production
- Seed investment for procurement entry and overseas expansion